



Hochschule München

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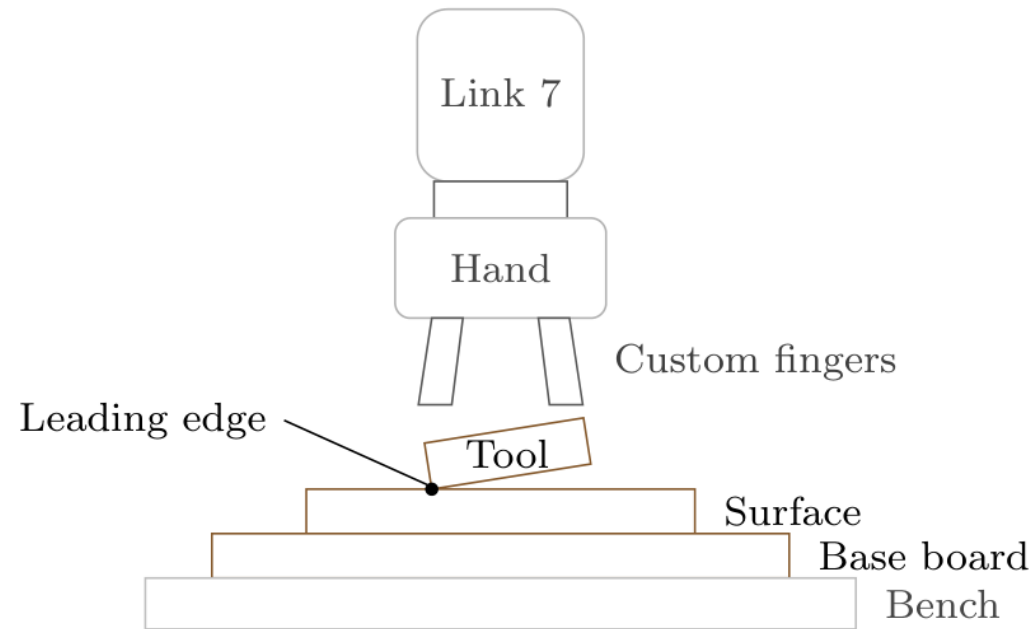
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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Problem

- Tilted tool → edge-first contact
- Rigid tracking resists corrective rotation

Main thesis question

- **How do Cartesian impedance settings affect passive alignment?**

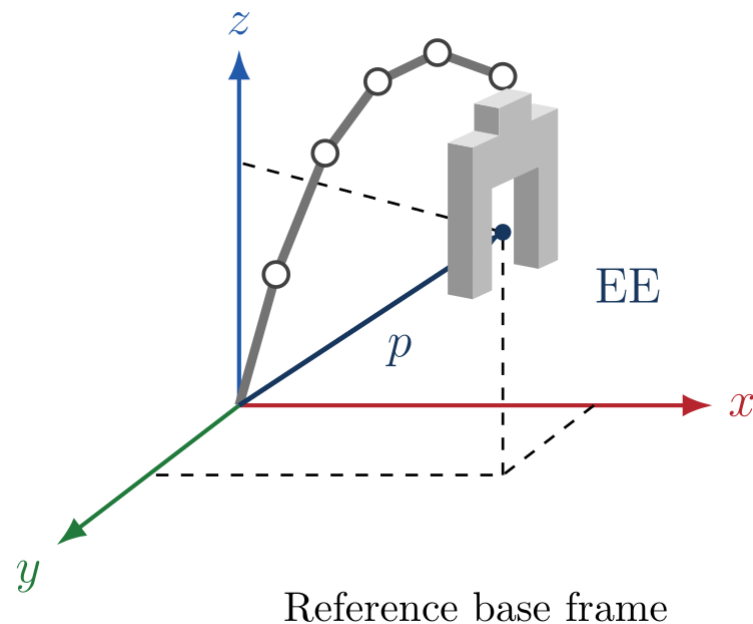
Idea

- Apply the press; let contact rotate the tool

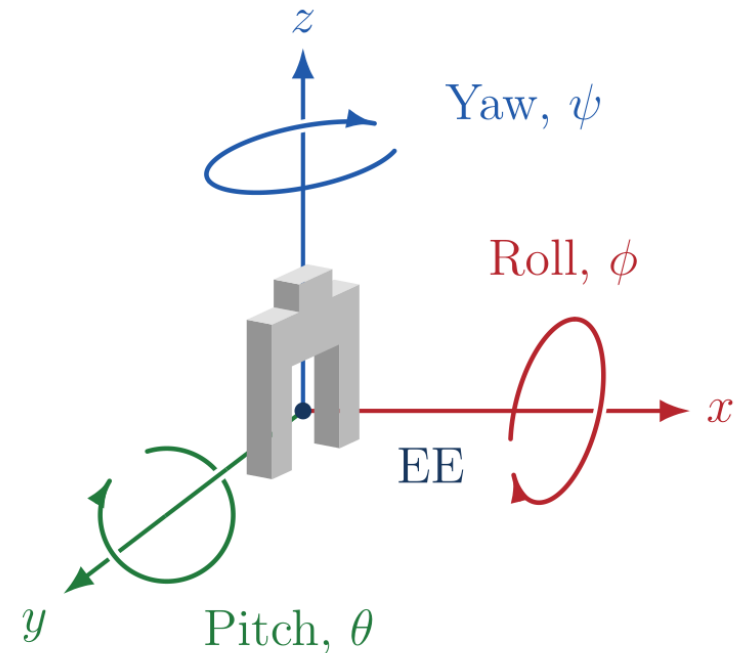
1. Introduction
2. Controller design
3. Experimental methodology
4. Validation
5. Experimental results
6. Conclusion and future work

In 3D, the end effector's position and orientation have 6 degrees of freedom.

Position: 3 translations



Orientation: 3 rotations



- The EE position depends on the configuration of all 7 joint angles: $p = p(q)$

- Position correction by force

$$e_p = p_d - p_{EE}$$

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

- Orientation correction by moment

$$e_R = \phi u, \quad R_d R_{EE}^\top = R(u, \phi)$$

$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

- Combined Cartesian wrench

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & \mathbf{0} \\ \mathbf{0} & K_R \end{bmatrix}}_{K_c} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & \mathbf{0} \\ \mathbf{0} & D_R \end{bmatrix}}_{D_c} \begin{bmatrix} \dot{p}_d - \dot{p}_{EE} \\ \omega_d - \omega_{EE} \end{bmatrix}$$

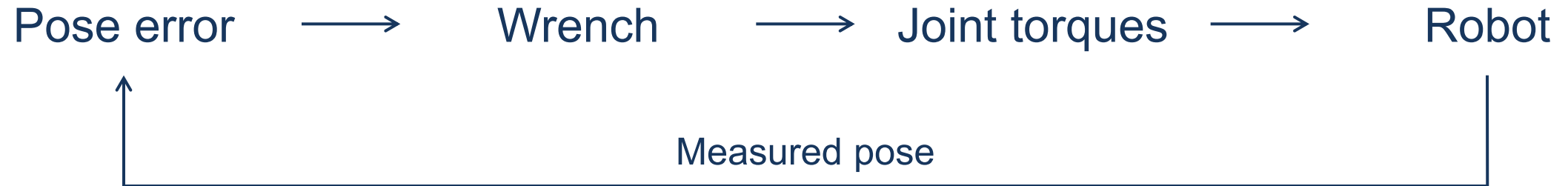
- Zero off-diagonal blocks: translation and rotation are decoupled.

K_p, K_R translational / rotational stiffness

D_p, D_R translational / rotational damping

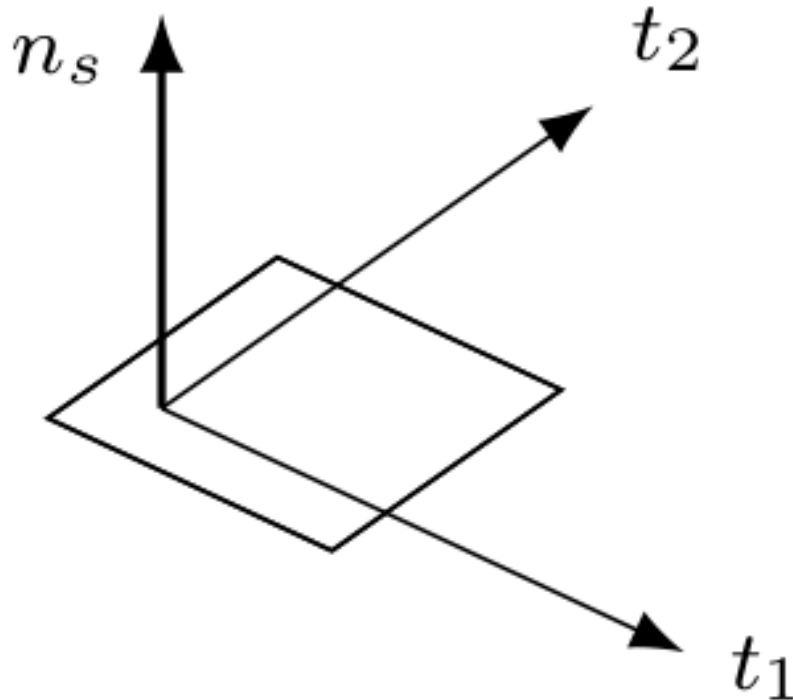
$\dot{p}, \omega$ linear / angular velocity

d, EE desired / measured end effector



$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \quad \tau_{\text{cart}} = J^{\top}(q)F$$

1 kHz control loop · C++ / libfranka



- n_s outward surface normal
- t_1, t_2 two perpendicular surface tangents

Translation

- Softer normal pressing
- Stiffer motion along the surface

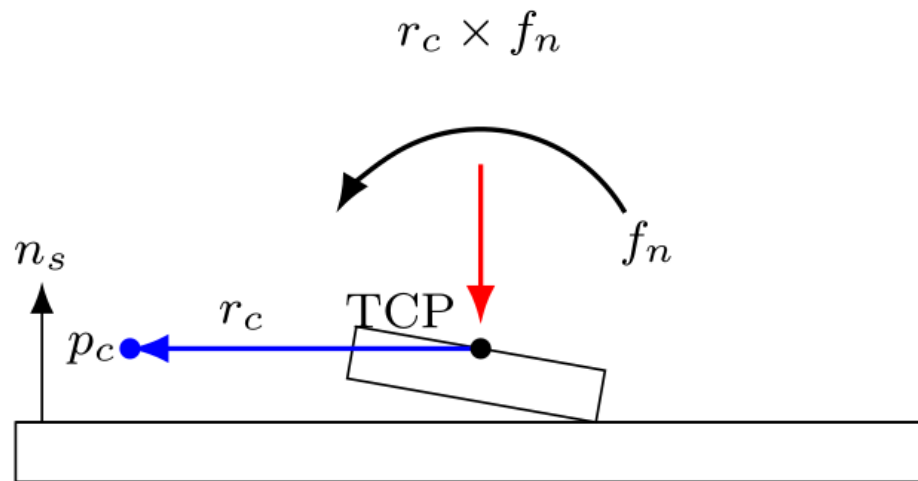
Rotation

- Softer rotations about the tangents
- Stiffer rotation about the normal

Virtual reference point

p_c CoC position

r_c Displacement from TCP to CoC

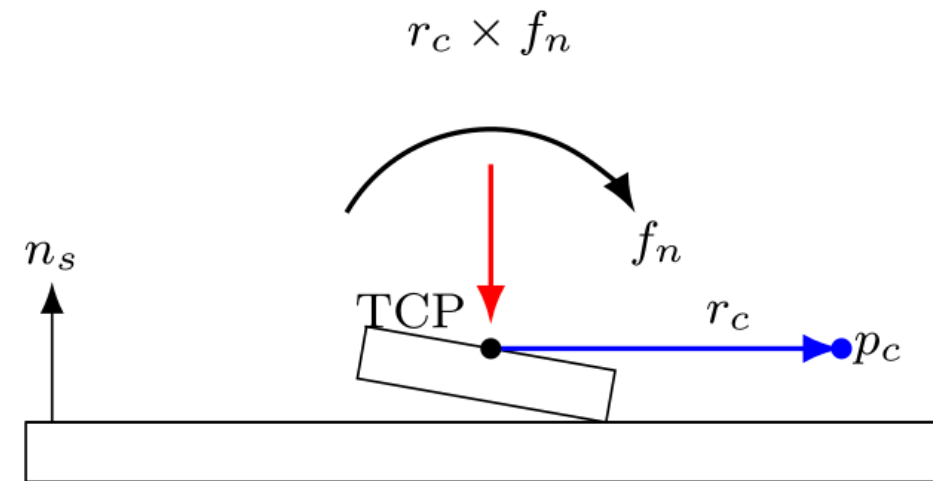


(a) Supporting displacement

Commanded TCP moment

$$m = m_R + r_c \times f$$

m_R rotational spring-damper moment



(b) Reversed displacement

- Opposite displacement reverses the added moment

- Shift from TCP to centre of compliance

$$A = \text{Ad}(r_c), \quad e_c = Ae$$

Local small-angle model; uncoupled stiffness and damping at the CoC

- Stiffness

$$K_{\text{TCP}} = A^\top K_c A$$

- Damping

$$D_{\text{TCP}} = A^\top D_c A$$

- A non-zero CoC displacement creates translation–rotation coupling.
- Translational error and velocity can contribute to commanded moment.

- Redundancy

7 joints for a 6-DOF Cartesian pose.
One redundant direction at full rank.

$$\dim(\ker J) = 7 - 6 = 1$$

- Cartesian task preservation

Zero instantaneous TCP translation and rotation

$$J(q) \dot{q}_{\text{null}} = 0$$

- Damping τ_d

Opposes redundant joint velocity.
Suppresses unwanted joint motion.

- Conditioning τ_σ

Seeks better Jacobian conditioning.
Can initiate joint motion.

- Singularity: a configuration where the Jacobian loses rank.

Tool orientation → Surface approach → Contact establishment → Hold

Compliant contact settings

Normal translation

$$K_{p,n} = 350 \text{ N/m}$$

Rotations about the surface tangents

$$K_{R,t_1} = K_{R,t_2} = 5 \text{ N m/rad}$$

Tangential translation

$$K_{p,t_1} = K_{p,t_2} = 2000 \text{ N/m}$$

$$\zeta = 1.0 \text{ inertia-scaled damping}$$

Controlled comparisons

- A: TCP baseline; zero, positive and negative entry tilt
- B: rotational stiffness about t_1
- C: translational stiffness along t_2
- D: tangential CoC displacement

19 settings × 3 trials = 57 contact trials

- 40 × 120 mm tool on a planar surface; fixed entry reference during 5 s contact.

Commanded joint disturbance

$$\tau_{\text{dist}} = J_p(q)^\top f_{\text{dist}}$$

Joint-torque equivalent of a virtual 20 N point force on link 3.

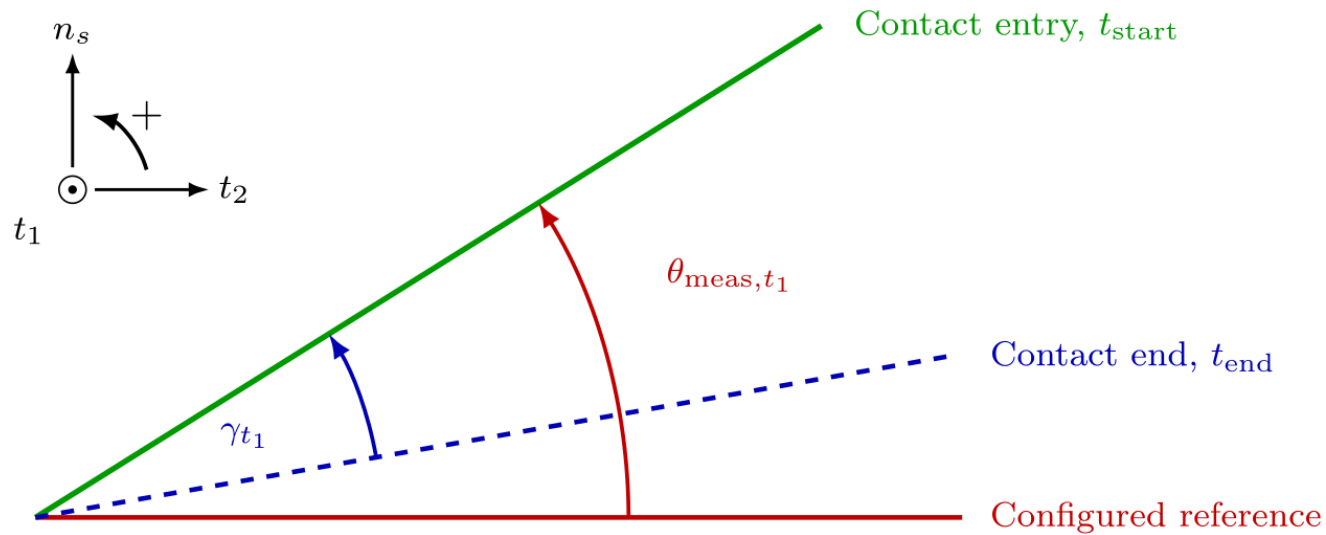
Cartesian pose hold

$$\|p_d - p_{\text{EE}}\| < 2 \text{ mm}$$

Hold the captured Cartesian pose. Assess TCP position retention.

Four settings; three trials per setting

- No null-space torque
- Damping: 2 N m s/rad
- Conditioning: 1.5 and 2.0 N m
- Compare accumulated motion and net displacement
- The disturbance excites joint motion while the Cartesian controller holds the TCP.

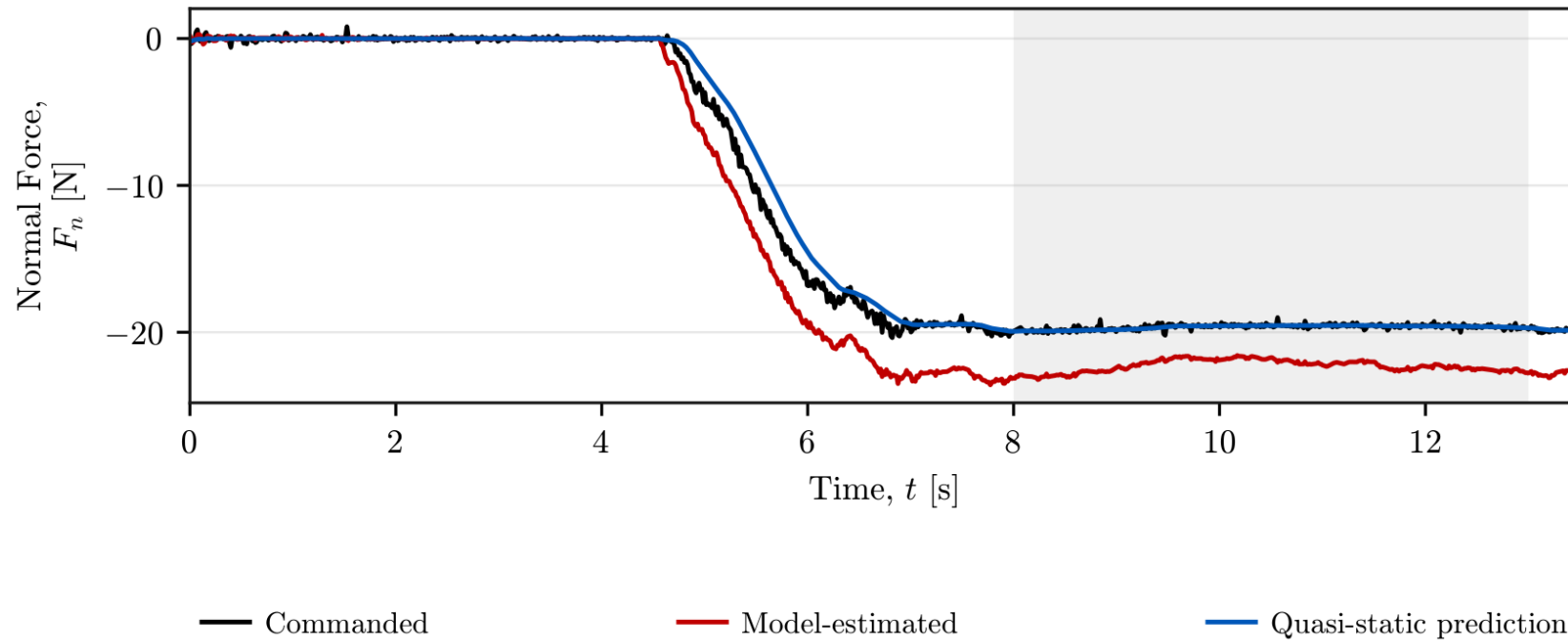


θ_{meas, t_1} Measured angular offset
Relative to the configured
surface reference

γ_{t_1} Contact response
Rotation from contact end
back to entry orientation

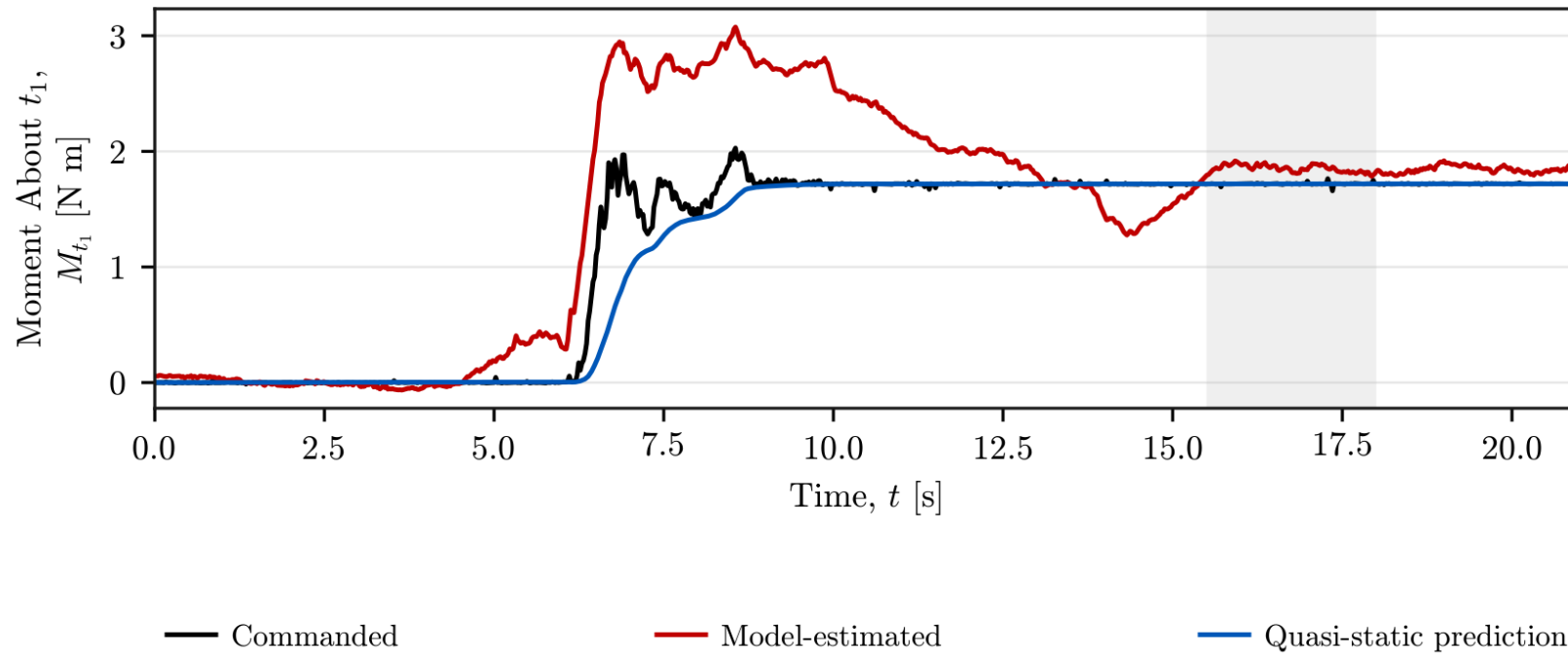
- Aligning response: same sign as the measured angular offset
- Calculated from measured end-effector orientation

$$F_{n,qs} = K_{p,n} n_s^\top (e_p - e_{p,0})$$



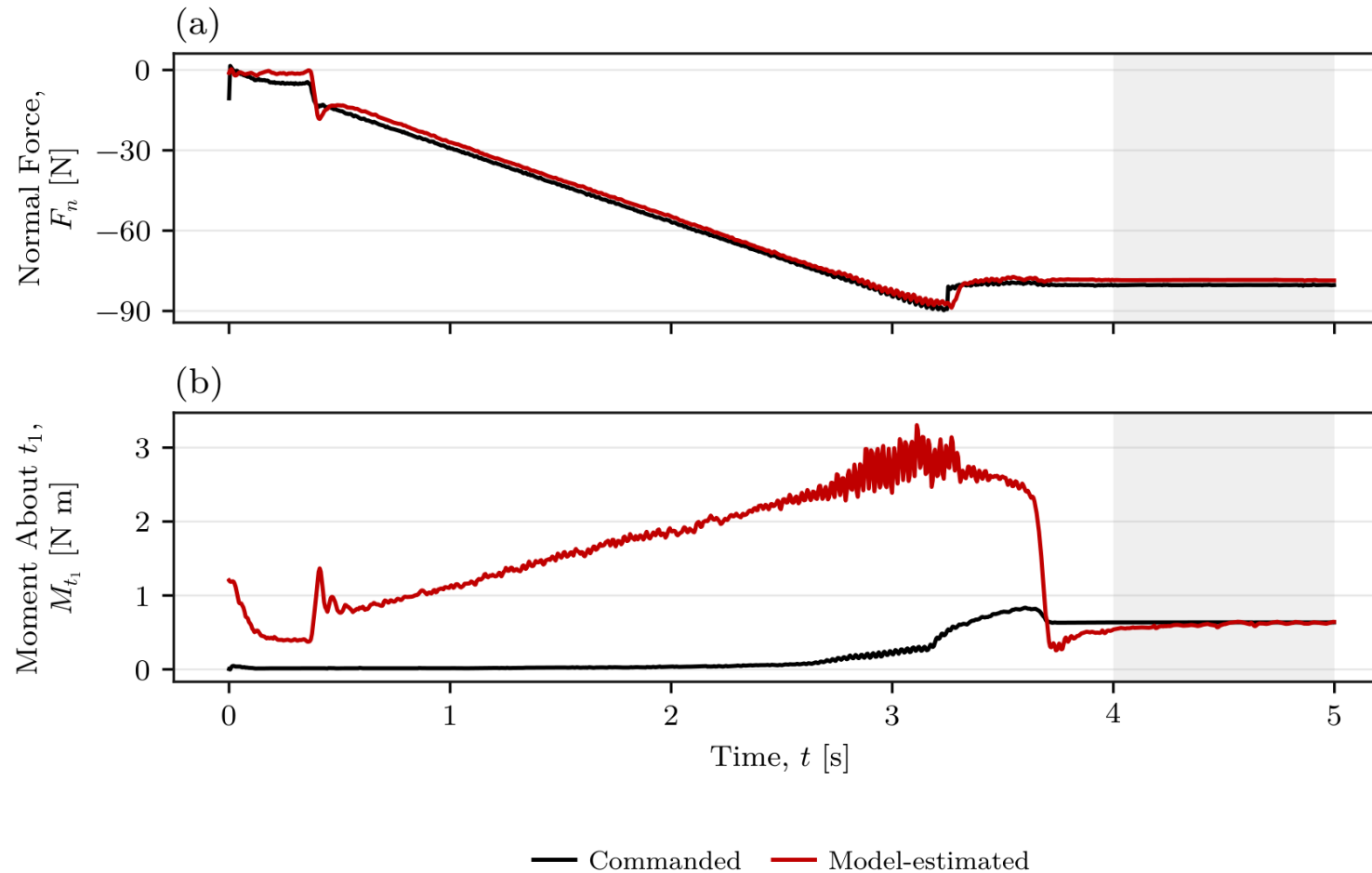
- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0})$$



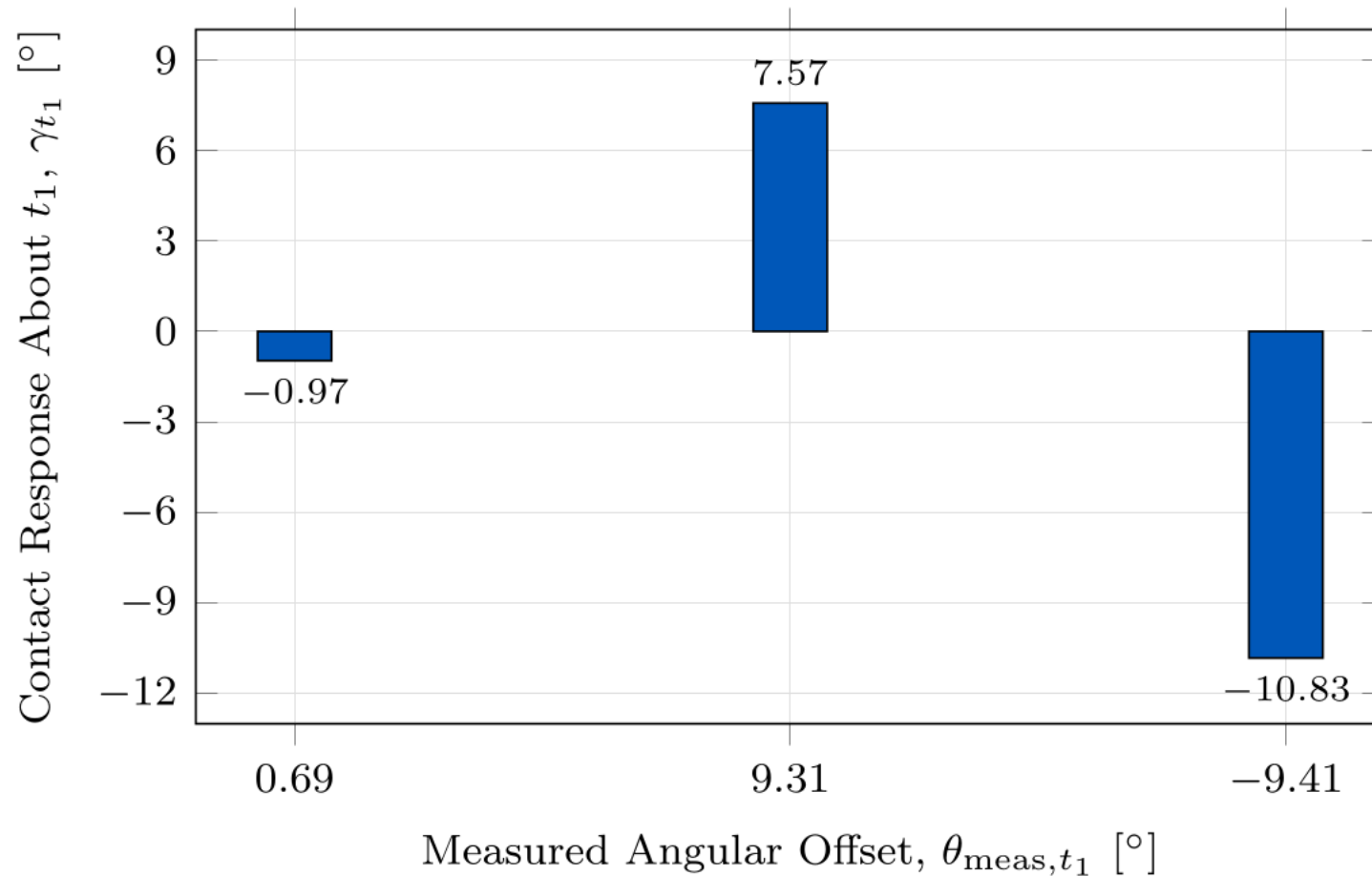
- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m

Commanded and estimated wrench



- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- TCP-centred trial; common axes and reference point

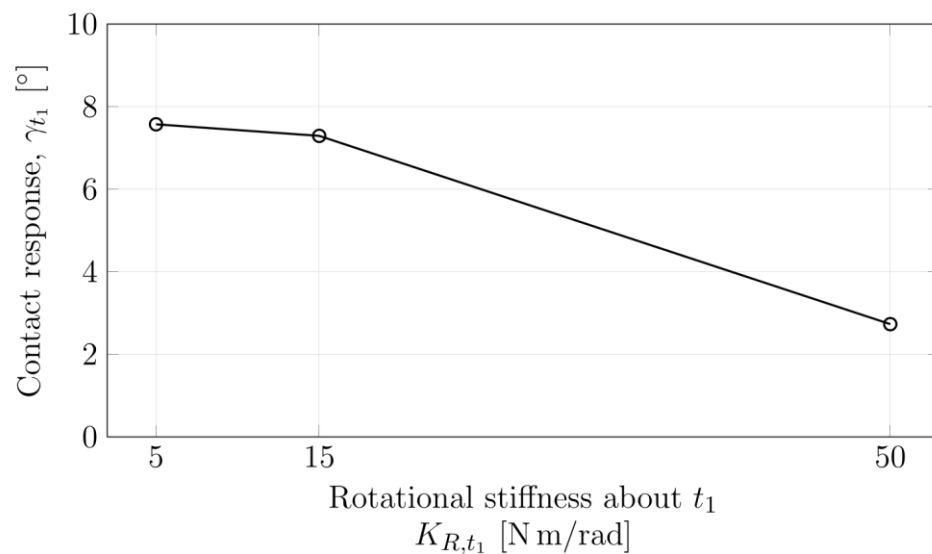
Baseline contact response



- Centre of compliance at the TCP
- Alignment response for both tilt signs
- Unequal response magnitudes

Rotational stiffness

Resistance to rotation about t_1

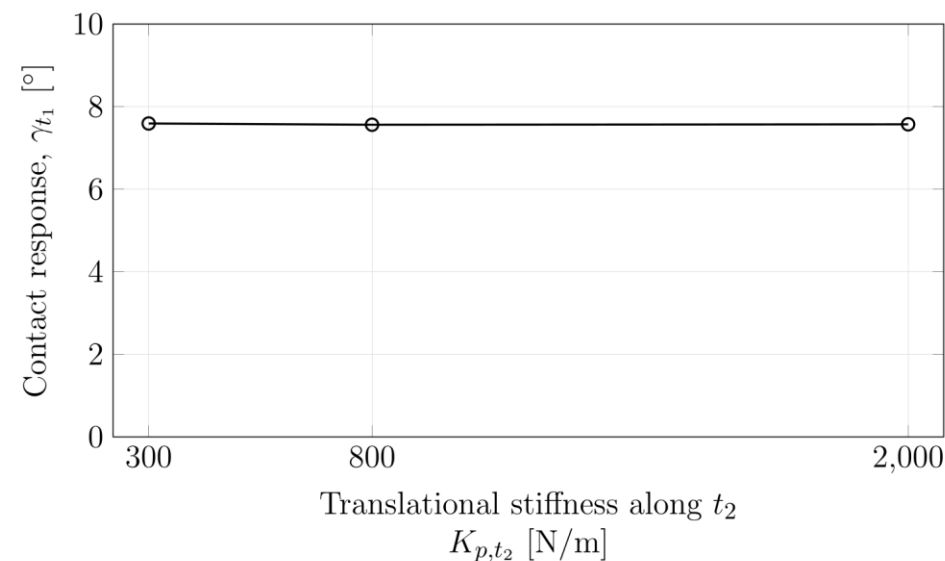


—○— Measured angular offset
 $\theta_{\text{meas},t_1} = (9.31\text{--}9.32)^\circ$

- Rotational stiffness: 63.9% lower response

Translational stiffness

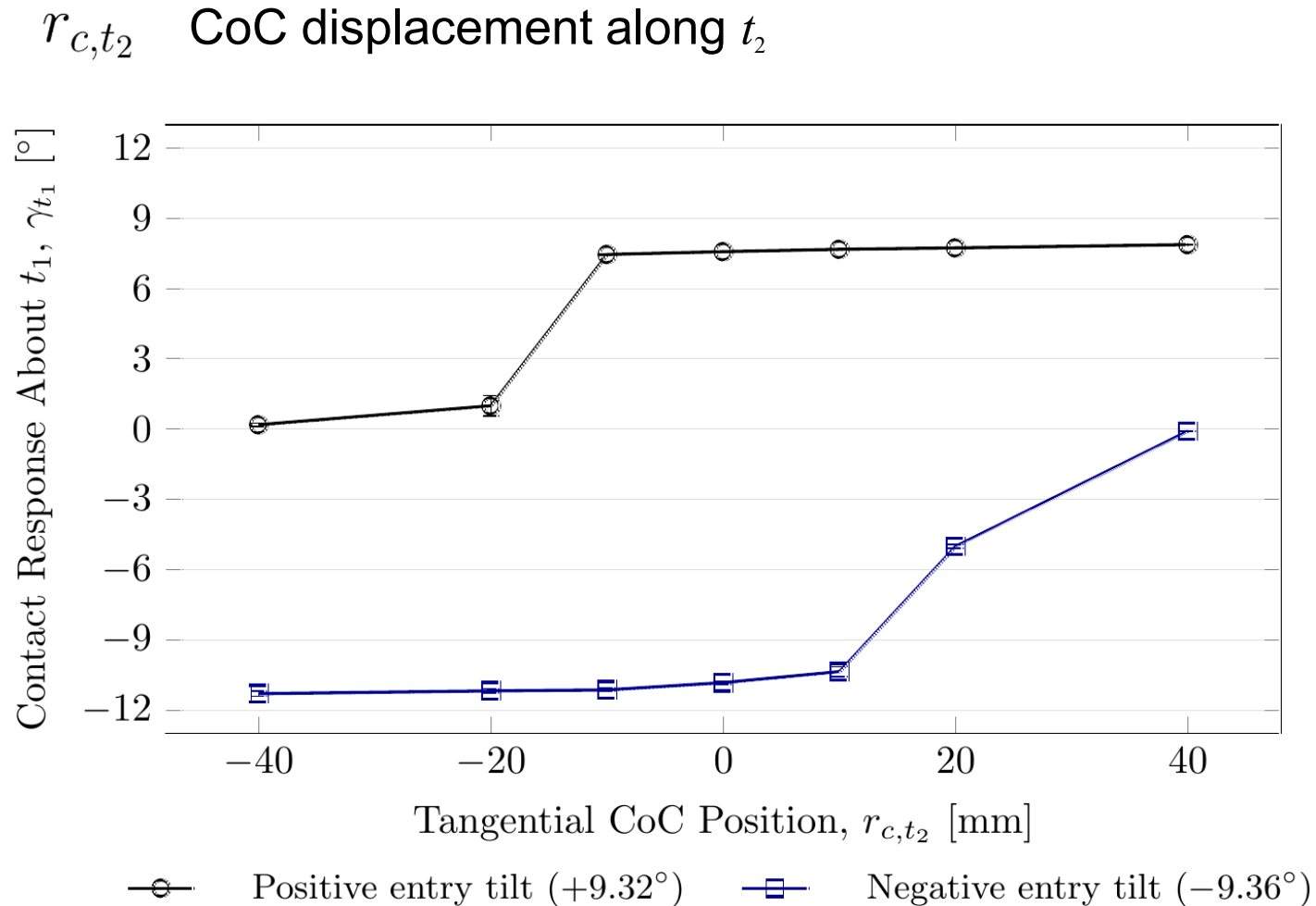
Resistance to sliding along t_2



—○— Measured angular offset
 $\theta_{\text{meas},t_1} = (9.24\text{--}9.31)^\circ$

- Tangential stiffness: 0.4% response span

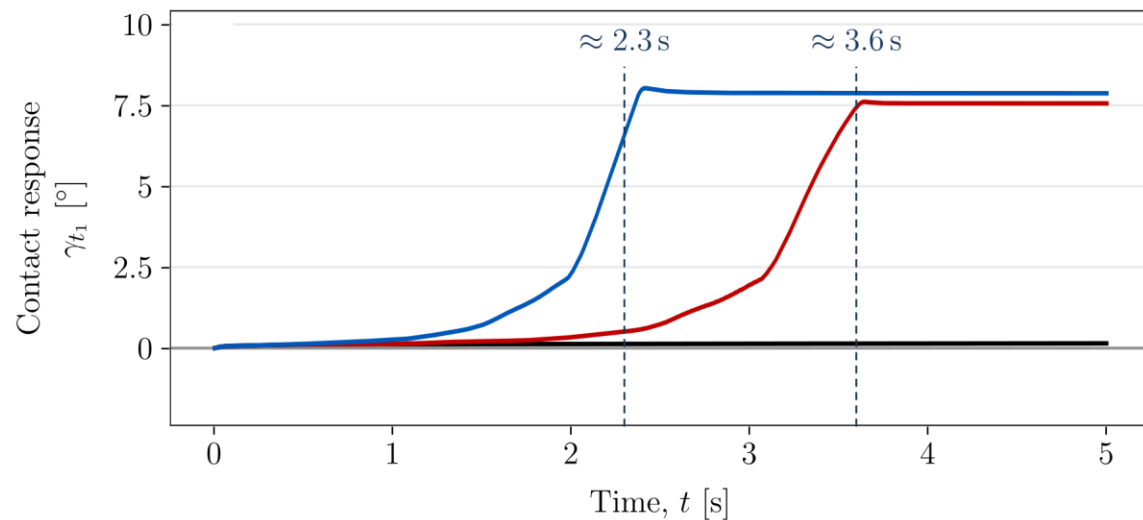
Effect of CoC position



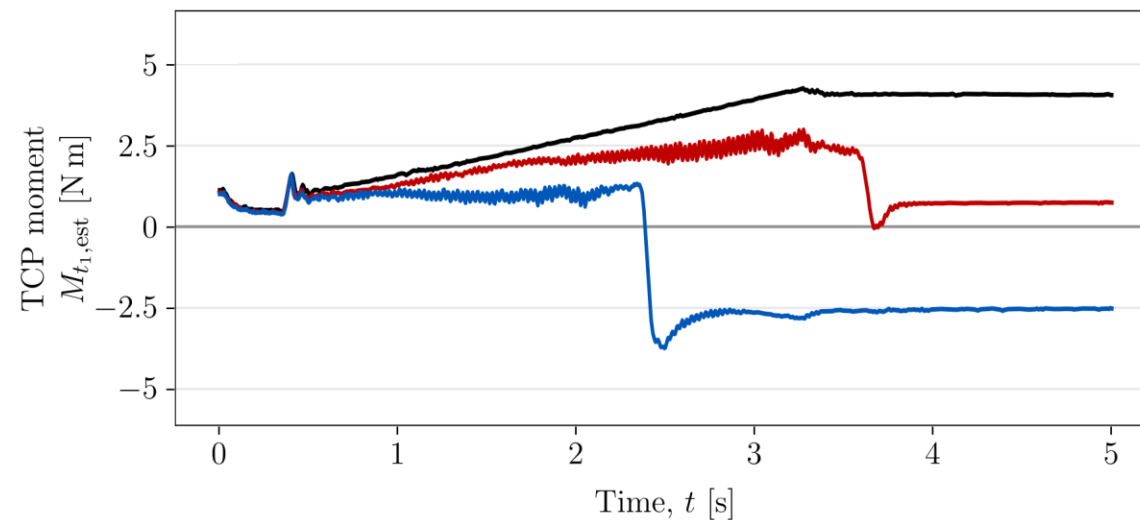
Relative to the TCP-centred baseline

- Supporting displacement: about 4% higher response
- Opposite side: 98–99% lower response
- Required side reverses with the entry tilt

Contact response

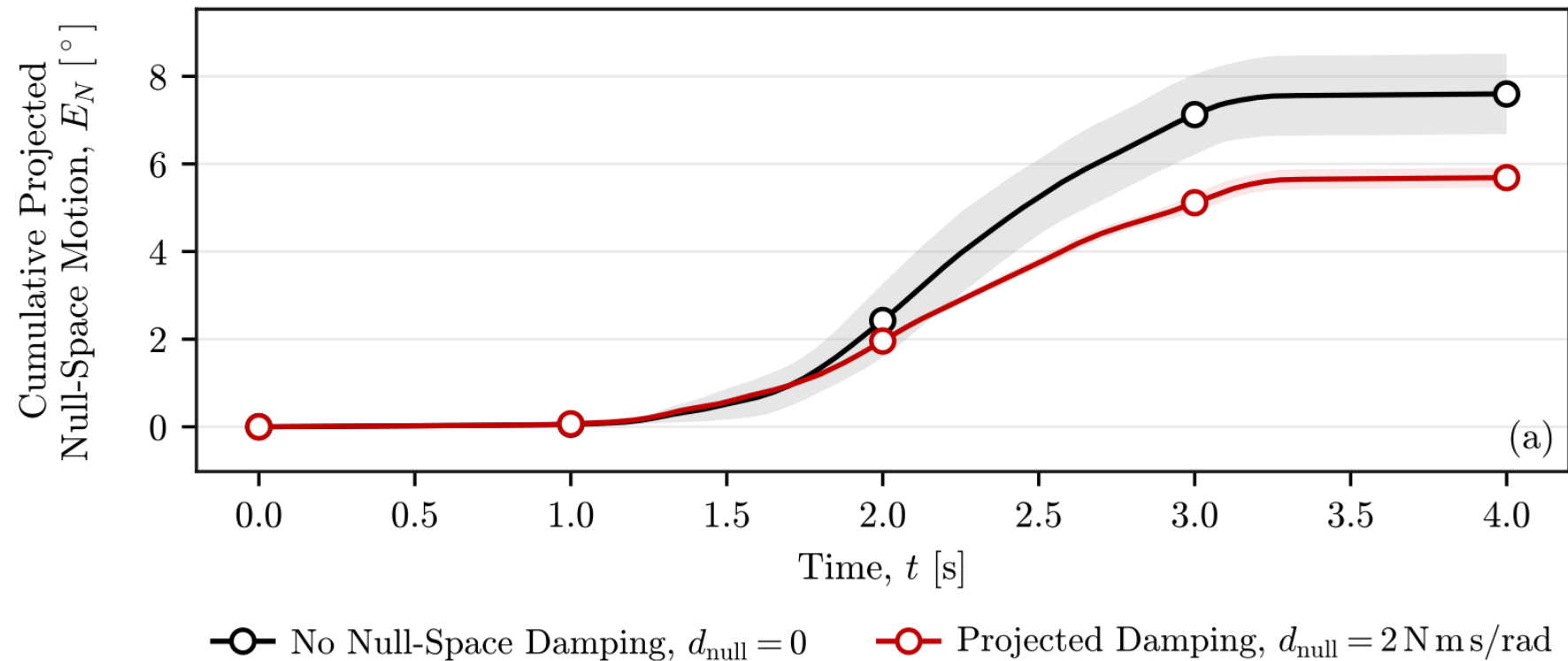


Interaction moment at the TCP

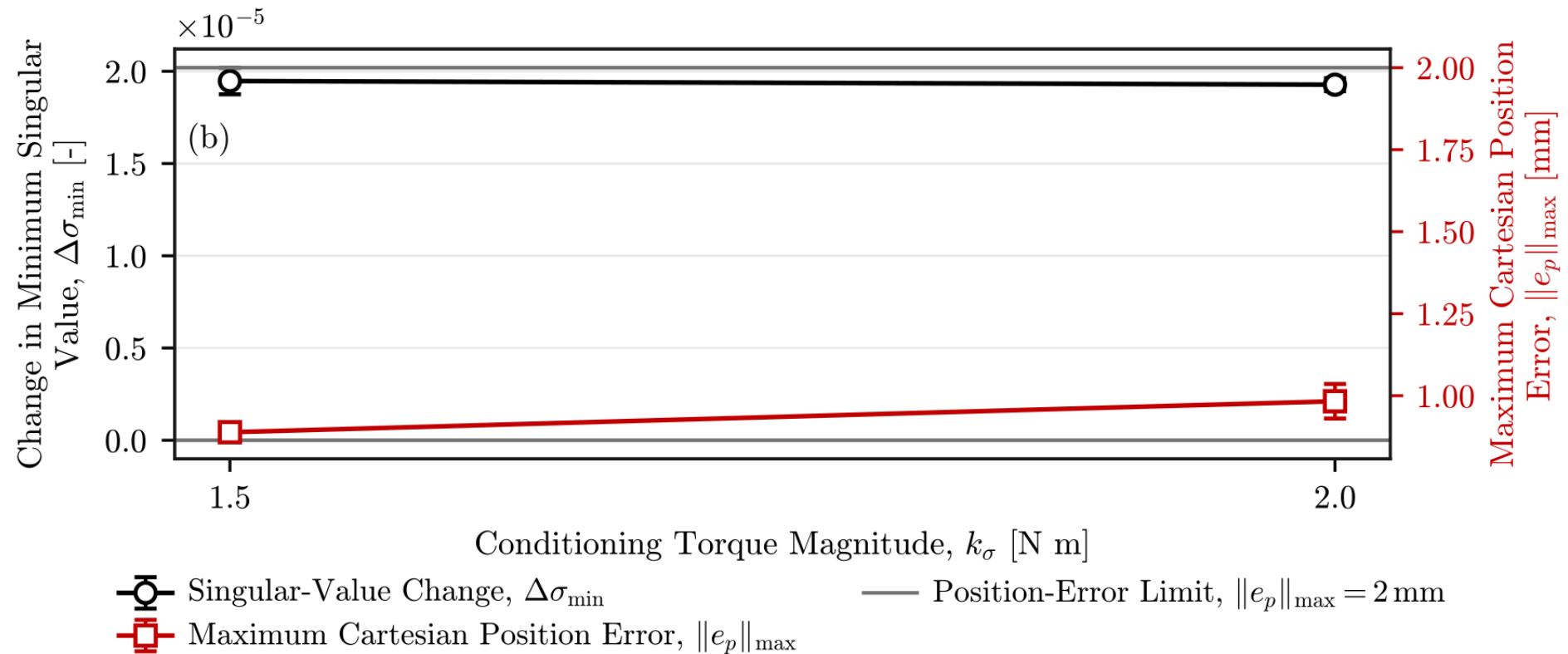


- Reversing CoC displacement reverses the interaction moment
- Supporting displacement: about 1.3 s earlier in these representative traces
- Final normal force (approx.): -83 N (-40 mm), -79 N (TCP), -78 N (+40 mm)

Null-space results

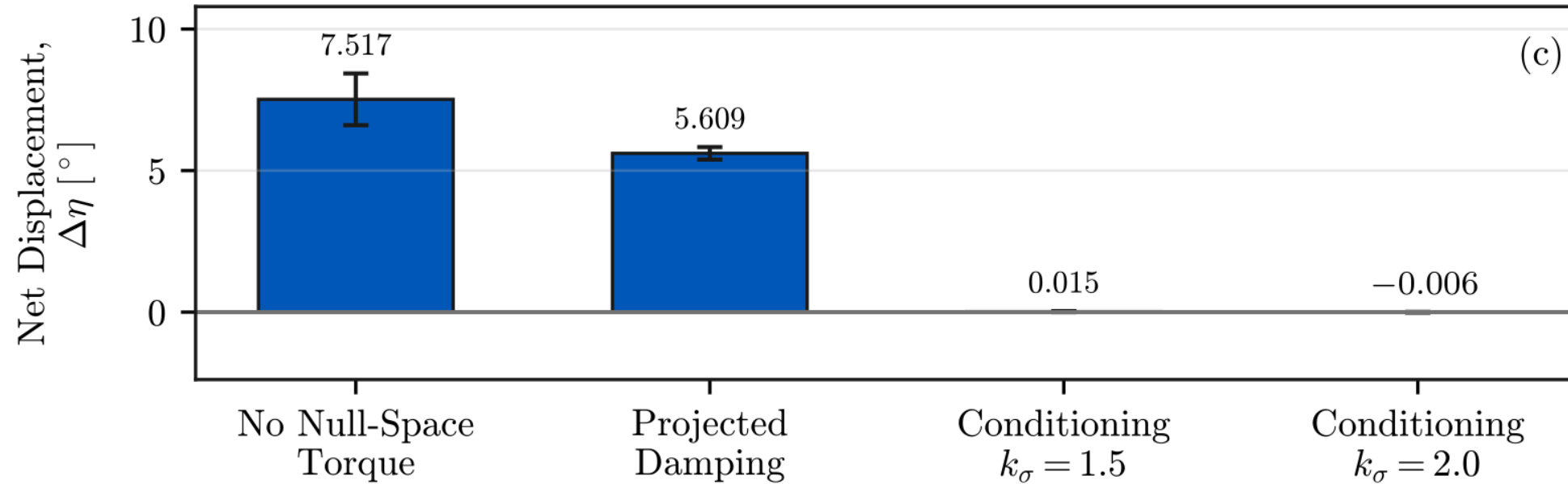


- Damping reduced cumulative motion from 7.596° to 5.687° (25%).
- Same commanded disturbance; three trials per setting.



- Both settings improved the local conditioning indicator.
- Maximum TCP error across all trials: 1.304 mm (criterion: 2 mm).

Net null-space displacement



- Conditioning reduced the net displacement to nearly zero.
- Small net displacement can coexist with back-and-forth motion.

Conclusion

- Higher rotational stiffness reduced passive contact-induced rotation.
- Tested t_2 translational stiffness had little effect; t_1 was fixed.
- CoC coupling supports or opposes rotation; the TCP is neutral.
- Null-space control changed configuration with TCP error below 2 mm.

Future work

- Stiffen the tool mount, especially for rotation about t_2 .
- Adapt CoC to entry tilt; return to the TCP after alignment.

The EE frame is attached to the controlled tool centre point (TCP).

Position

$$p_{EE} \in \mathbb{R}^3$$

3 translational degrees of freedom
TCP position in the robot base frame

Orientation

$$R_{EE} \in SO(3)$$

3 rotational degrees of freedom
EE rotation relative to the robot base

Measured pose

$$T_{EE} = \begin{bmatrix} R_{EE} & p_{EE} \\ 0^\top & 1 \end{bmatrix}$$

Obtained directly from the robot state for position and orientation feedback.

7 joint coordinates control a 6-DOF tool pose: 3 position + 3 orientation.

$$\dim(\ker J) = 7 - \text{rank}(J) = 7 - 6 = 1$$

At a nonsingular configuration: one independent redundant motion direction.

This freedom is a coordinated motion of the joints, not one designated joint.

$$J(q) \dot{q}_{\text{null}} = 0, \quad J \in \mathbb{R}^{6 \times 7}$$

J maps joint velocity to tool velocity: null-space motion gives zero tool velocity.

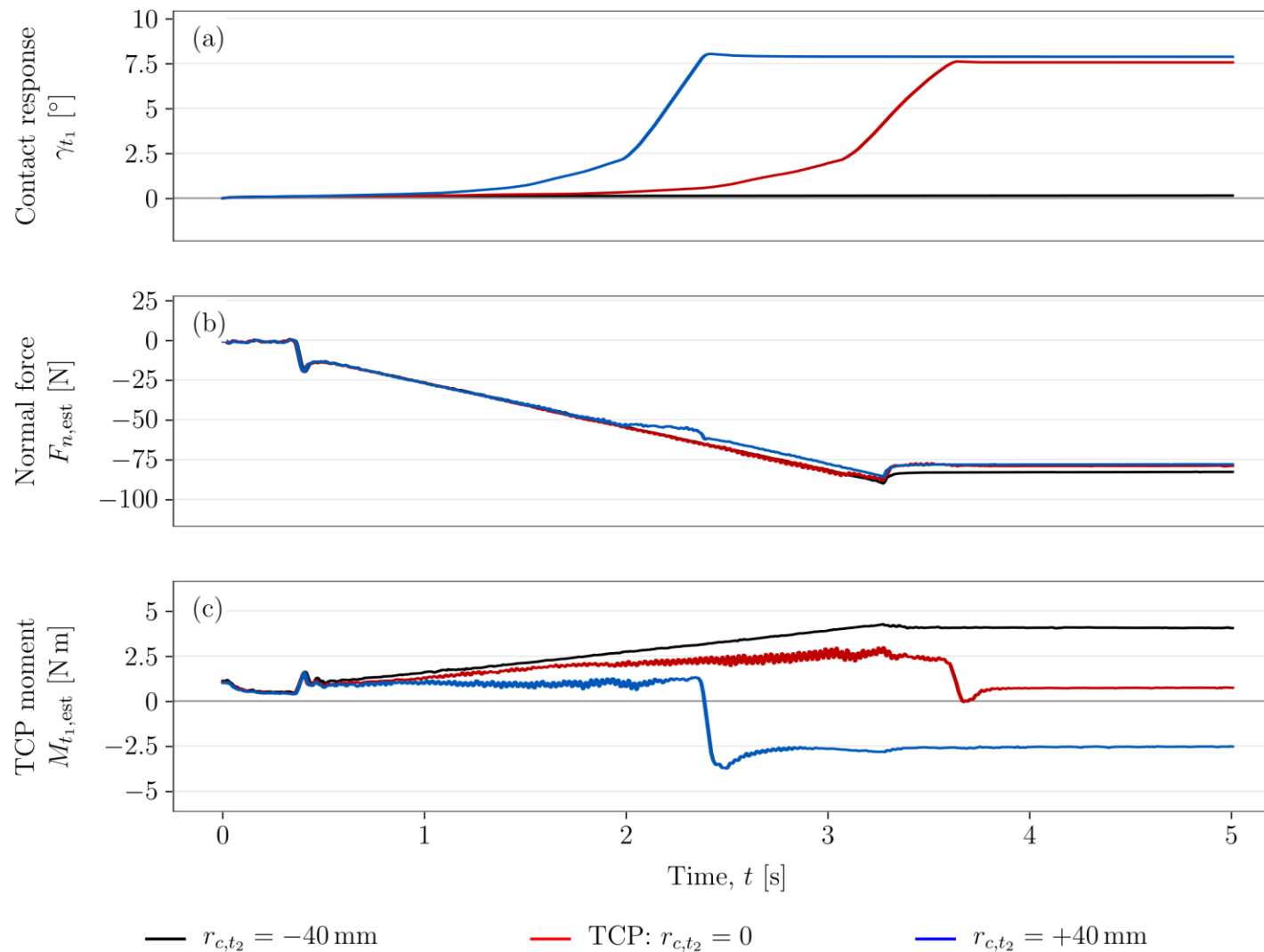
Two torque components control this internal motion:

$$\tau_{\text{null}} = \tau_d + \tau_{\sigma}$$

Damping torque: counters null-space joint motion.

Conditioning torque: steers towards a configuration with better motion capability.

Contact response, normal force and moment



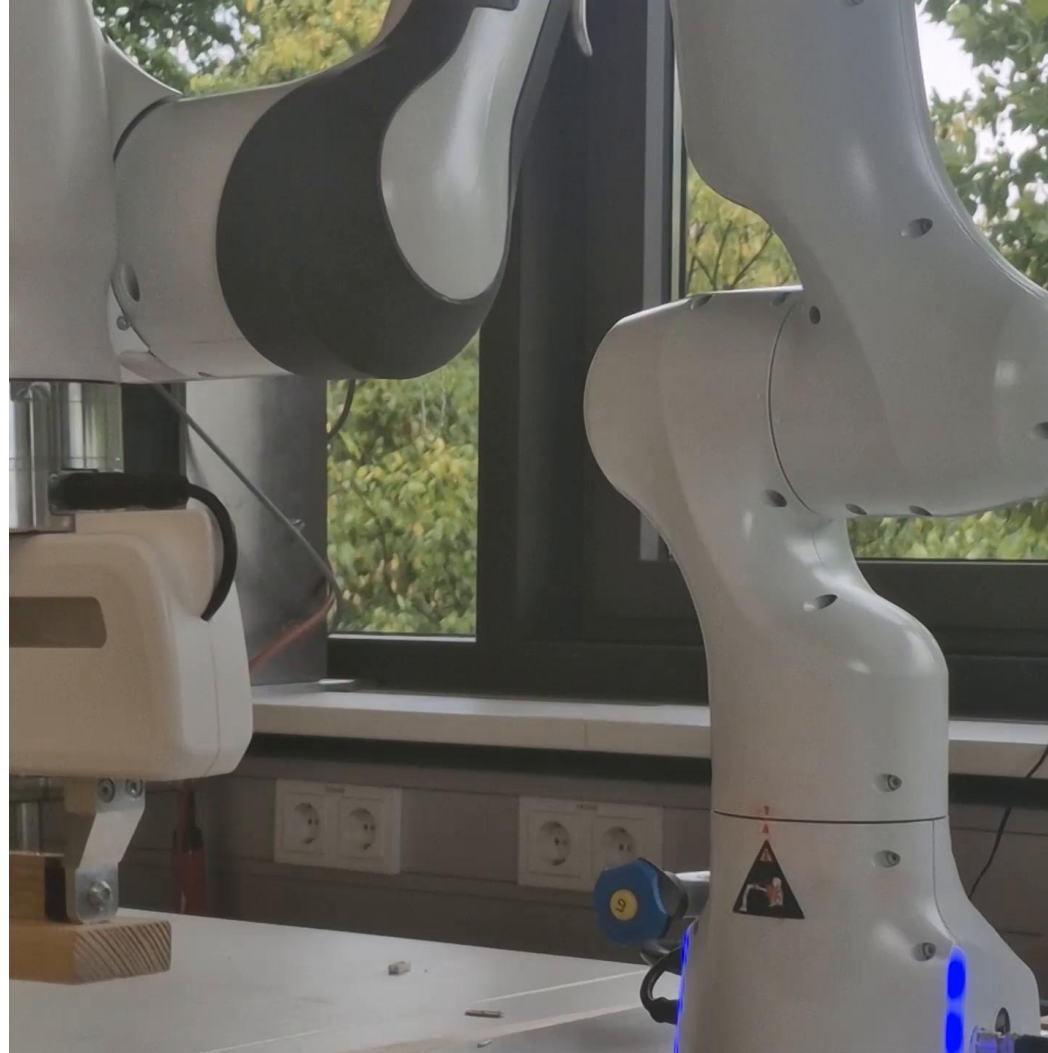
Representative traces
Positive entry tilt

- Similar final normal force
- Moment reverses with CoC displacement
- Force and moment: model-based robot estimates



Pre-grinding hold · fixed desired orientation · CoC at TCP

Disturbance demonstration



No null-space torque → Damping → Conditioning